

PC-08 · PRESERVATIVES & CHELANTS · PERSONAL CARE

# HS BHT

BHT

|      |          |           |        |                       |
|------|----------|-----------|--------|-----------------------|
| INCI | CAS      | EC        | FAMILY | SUPPLIED AS           |
| BHT  | 128-37-0 | 204-881-4 | PC-08  | Powder / flakes, ≥99% |

01 POSITIONING

The cheapest insurance policy a formula can carry — pennies against a recall.

02 HOW IT WORKS

- 01 Radical chain-breaker

Donates a hydrogen to intercept the lipid-oxidation chain before it propagates.
- 02 Formula protection

Guards oils, fragrance and vitamins from going rancid or discoloring on shelf.
- 03 Trace-level efficiency

Effective at 0.01–0.1% — negligible cost against the shelf-life it buys.

03 APPLICATIONS

- Oil-rich creams and balms
- Fragrance and flavor protection
- Color cosmetics
- Soaps and surfactant pastes
- Industrial and polymer stabilization

04 FORMULATION GUIDE

|                                                                                                   |                           |
|---------------------------------------------------------------------------------------------------|---------------------------|
| USE LEVEL                                                                                         | 0.01–0.1%                 |
| PH & CONDITIONS                                                                                   | Oil phase; pH-independent |
| — Dissolve in the oil phase during heating; a chelant alongside BHT covers both oxidation routes. |                           |
| — Do not over-dose — high levels can discolor and hit regional food-contact limits.               |                           |

05 TYPICAL SPECIFICATION

|               |                                   |
|---------------|-----------------------------------|
| Appearance    | White crystalline powder / flakes |
| Assay         | ≥ 99%                             |
| Melting point | 69–72 °C                          |
| Grade         | cosmetic / food per TDS           |

Typical parameters. The specification on the current TDS and the per-lot Certificate of Analysis govern.

06 PERFORMANCE — DIRECTIONAL

Oxidation induction time (class data)

With BHT

Cost vs shelf-life gained

BHT insurance

Values are representative of the ingredient class, based on published technical literature, and shown directionally for comparison. Final performance must be validated in the customer's formulation.